

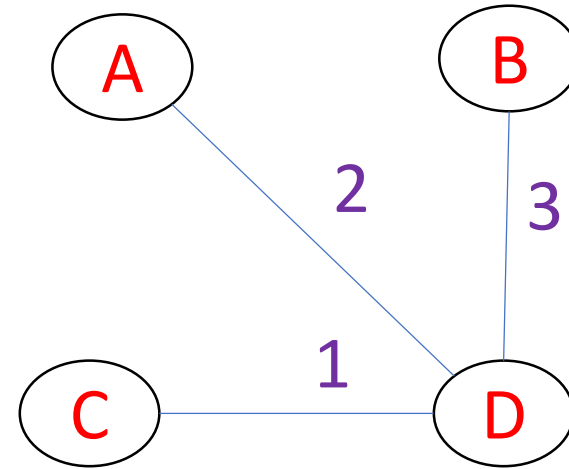
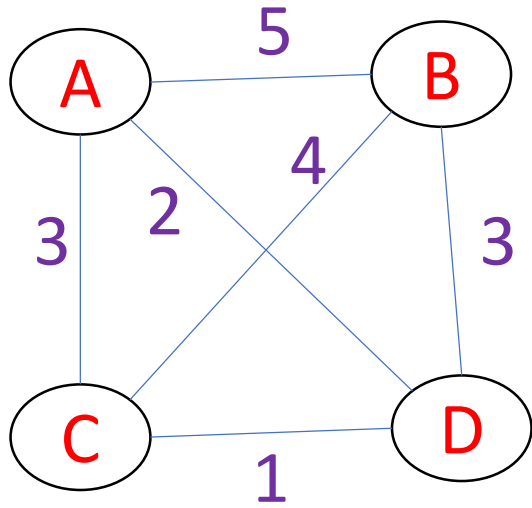


Krushkal's Algorithm

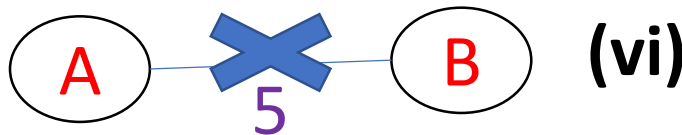
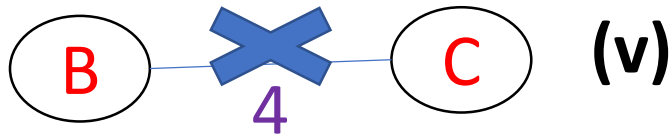
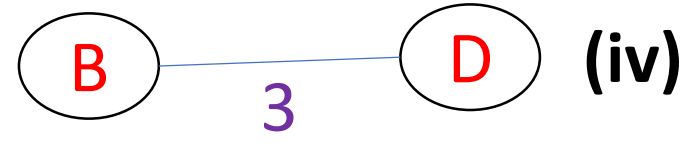
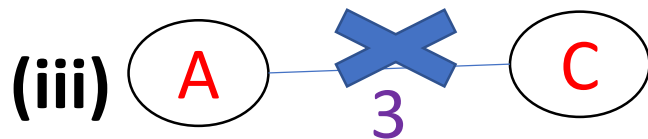
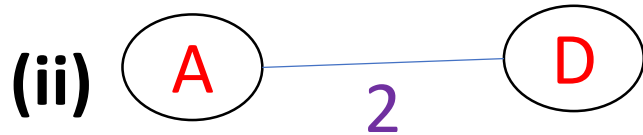
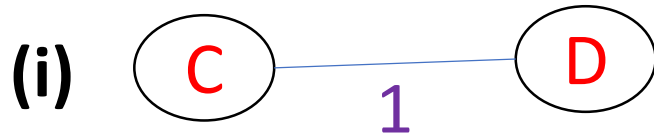
Minimum Cost Spanning Trees-Krushkal's Algorithm

- 1. Sort all the edges in increasing order of their weight.**
- 2. Pick the smallest edge. Check if it forms a cycle with the spanning tree formed so far. If cycle is not formed, include this edge. Else, discard it.**
- 3. Repeat step#2 until there are $(V-1)$ edges in the spanning tree.**

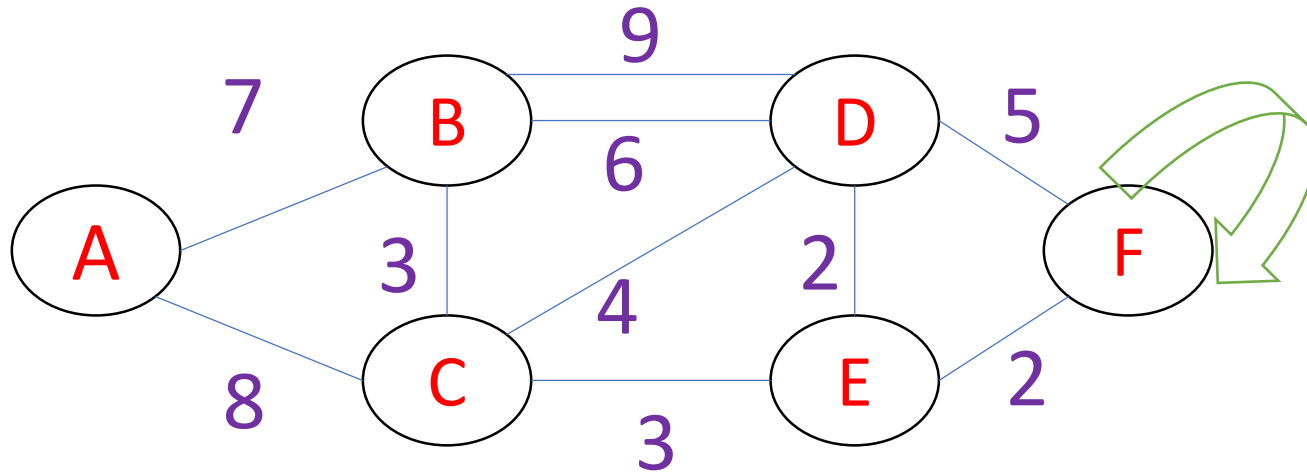
Minimum Spanning Trees-Krushkal's Algorithm



MST: 6



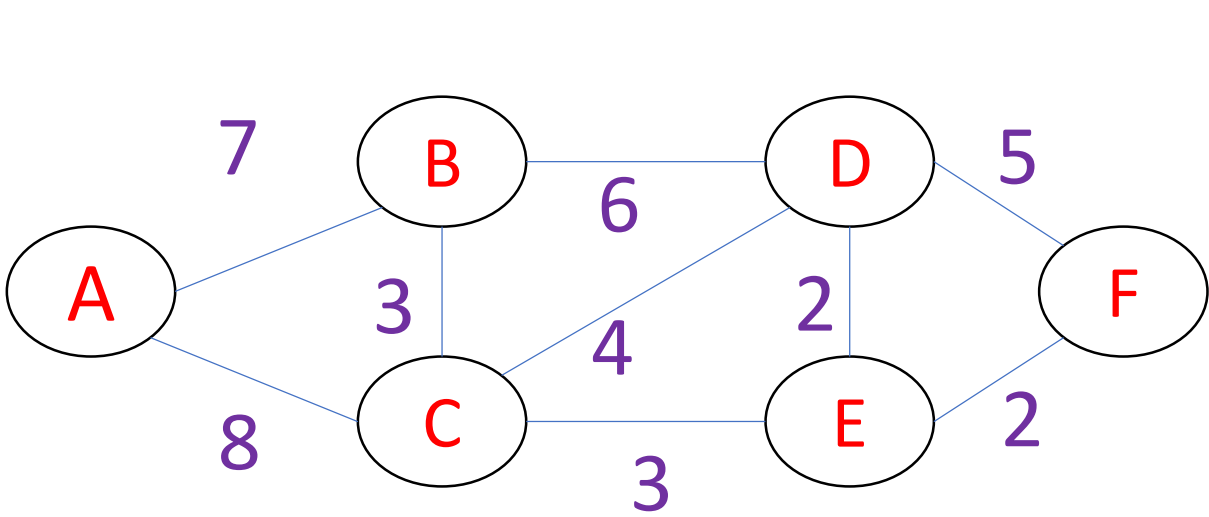
Minimum Spanning Tree Krushkal's Algorithm Example



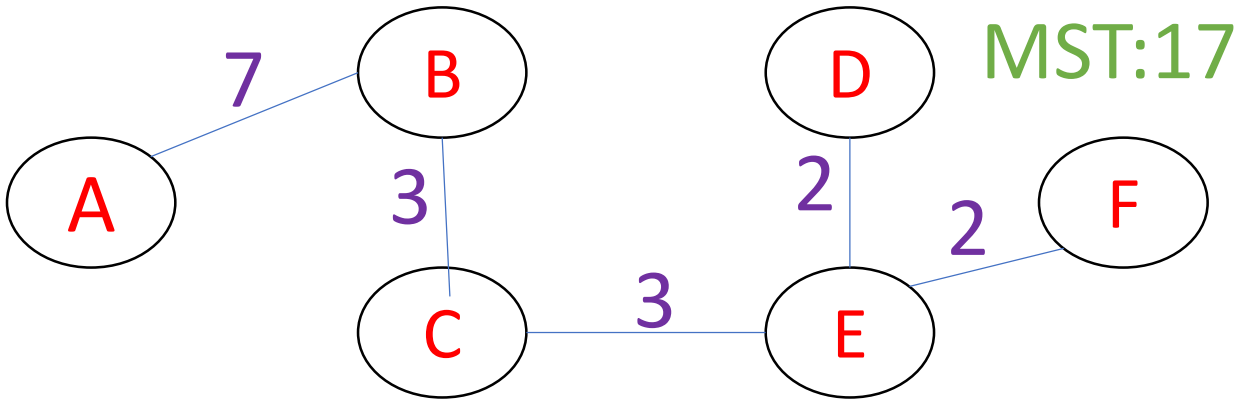
Timing Analysis of Krushkal's Algorithm

In **Kruskal's algorithm**, most time consuming operation is sorting because the total complexity of the Disjoint-Set operations will be $O(E \cdot \log V)$, which is the overall Time Complexity of the algorithm.

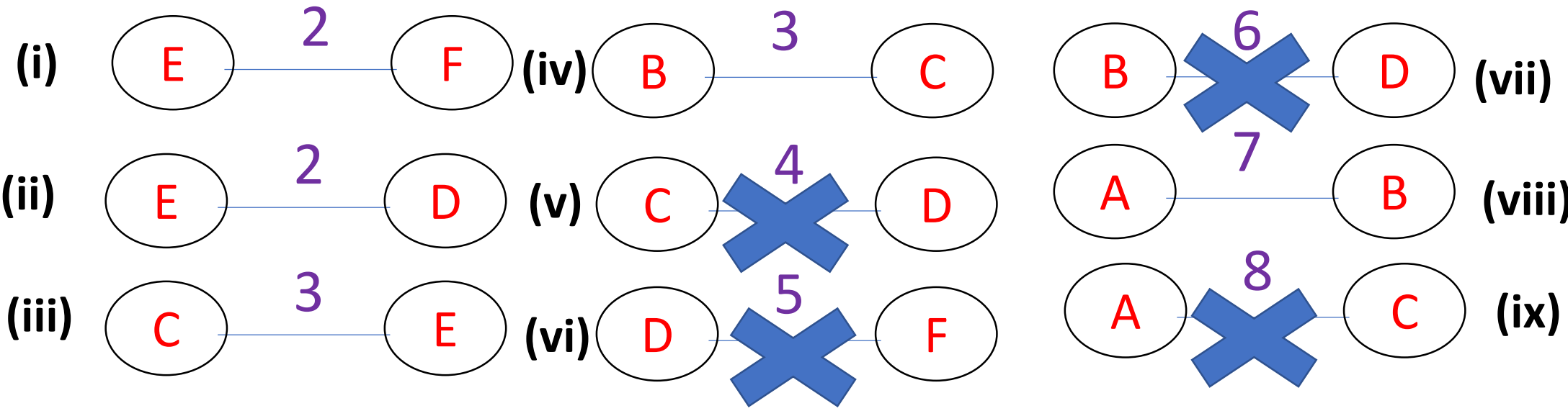
Minimum Cost Spanning Tree Krushkal's Algorithm

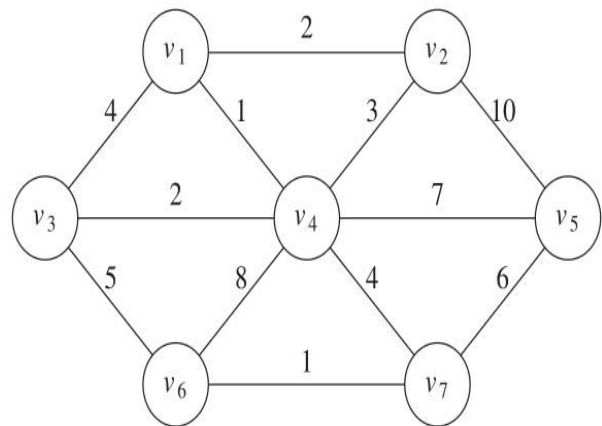


Graph G

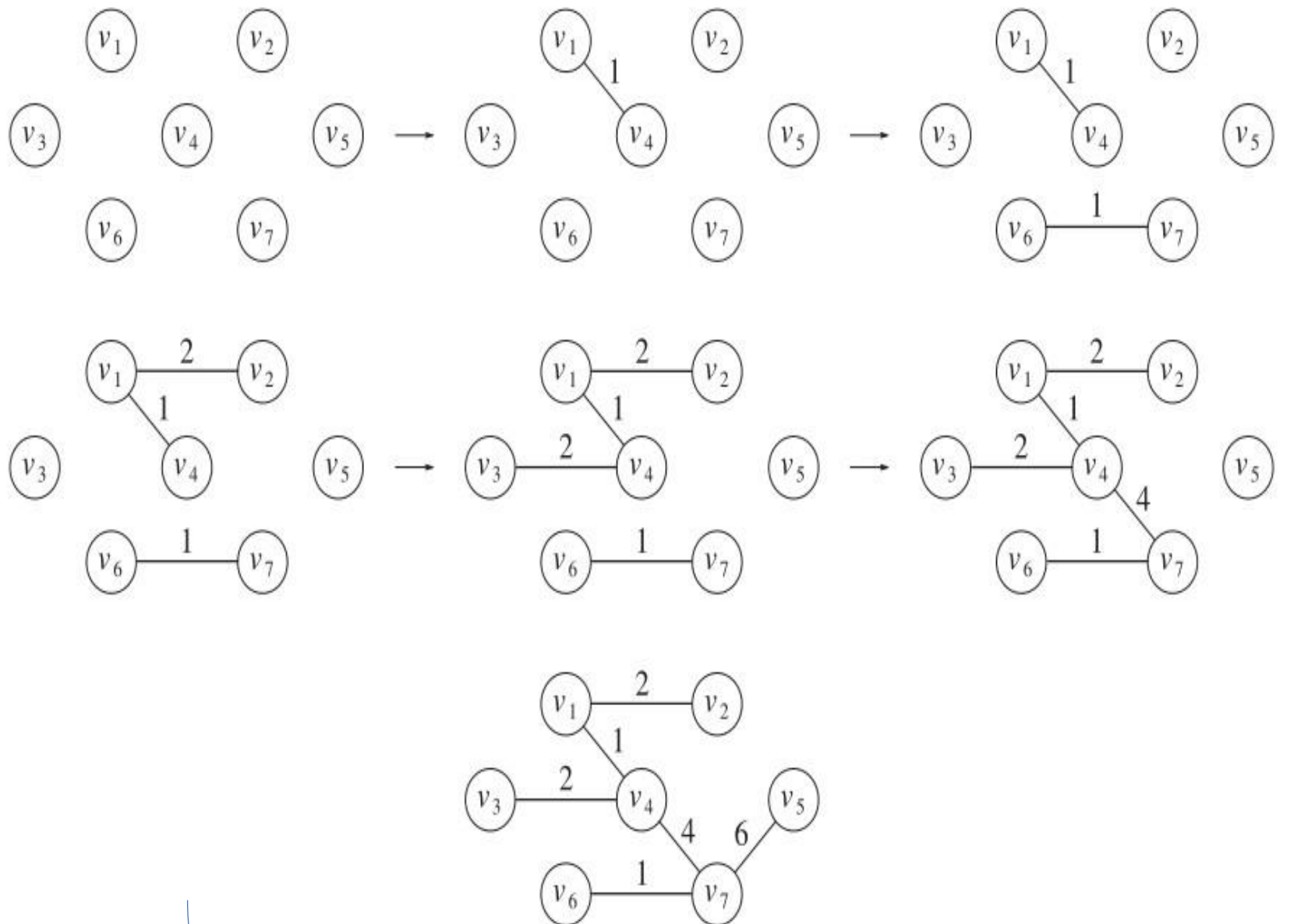


- 1. In the given graph G, delete all the loops and parallel edges. [delete with maximum cost]
- 2. Arrange all the edges in incremental order





Graph G



Spanning Trees